

Fitbit Tracker Data

1. Background

Wearable fitness trackers capture continuous, device-level signals on activity, sleep, and heart rate, alongside self-reported profile attributes such as occupation, lifestyle category, and dietary preference. On their own, these streams describe individual behaviour. Combined, they can reveal how different user segments actually engage with a wellness device day to day, where that engagement is strong, and where it drops off.

This is the same class of problem faced by consumer health-tech and wellness product teams: raw tracker exports are detailed but fragmented across activity, sleep, heart rate, and step logs, collected at different levels of granularity (daily, hourly, and minute-level) and keyed only by a user identifier. Before any pattern can be trusted, the data has to be modelled correctly and read at the right grain.

2. Problem Statement

A wellness product team has access to device usage data from a panel of users who vary in occupation, lifestyle category, self-declared fitness level, and diet type. The team wants to understand how activity, sleep, and heart rate patterns differ across these segments, in order to identify which groups are meeting healthy activity and sleep benchmarks, which are at risk of disengagement, and where a personalised intervention (reminders, goal recalibration, or content nudges) would be most useful.

The analysis is expected to answer four questions:

- How does daily activity (steps, active minutes, calories) vary by lifestyle category and occupation?
- Is there a measurable relationship between sleep duration or sleep efficiency (time asleep versus time in bed) and next-day activity levels?
- Which segments show activity or logging patterns consistent with declining engagement over the observed period?
- Do self-declared fitness level and diet type align with the activity and calorie data actually recorded, or is there a gap between stated and observed behaviour?

The intended output is a small set of views built on a unified data model, one that relates every measurement table back to the user profile table without duplicating rows across the very different grains (daily, hourly, and minute-level) that the source files use.

3. Data Sources

The dataset consists of five files sharing a common user identifier. Each file sits at a different level of granularity, summarised below.

File	Grain	Description
Fitbit_Users	One row per user	User profile: lifestyle category, self-declared fitness level, occupation, name, and diet details.
Fitbit_Activities	One row per user per day	Daily summary of steps, distance, activity intensity, and calories burned.
Fitbit_Sleep	One or more rows per user per day	Daily sleep sessions, including time asleep and time in bed.
Fitbit_Steps	One row per user per day per hour	Hourly step counts within each day.
Fitbit_HR	One row per user per date per hour per minute	Minute-level heart rate readings, the most granular table in the set.

Because the tables differ in grain, they are related to Fitbit_Users on the shared identifier rather than joined directly to one another, which would multiply rows incorrectly across mismatched levels of detail.

4. Data Dictionary

4.1 Fitbit_Users

Field	Description
Id	Unique identifier for the user. Primary key of this table; referenced by all other files.
Category	Lifestyle segment, e.g. Busy Mum, Active Millennial, Casual Exerciser.
Self-declared fitness form	User's own rating of their fitness level, e.g. Good Fitness, Top Fitness, Average, Out of Shape.
Occupation	Stated occupation. Not populated for every user.
Gender	User's gender, parsed from the combined Gender/age field.
Age	User's age in years, parsed from the combined Gender/age field.
First name / Middle name initial / Last name	Name components as recorded in the source file.
Max Calories	Daily calorie target, parsed from the combined diet details field.
Diet Type	Stated dietary preference, e.g. Keto, Paleo, Vegan, No Restrictions, parsed from the combined diet details field.

4.2 Fitbit_Activities

Field	Description
User_ID	User identifier. Foreign key to Fitbit_Users.Id.
ActivityDate	Calendar date of the activity record.
Steps	Total steps recorded for the day.
Distance	Total distance covered for the day.
Fast KM	Distance covered at a fast pace, in kilometres.
Slow KM	Distance covered at a slow pace, in kilometres.
Active Min	Minutes classified as active during the day.
Inactive Min	Minutes classified as inactive (sedentary but awake) during the day.
Rest Min	Minutes classified as resting during the day.
Calories	Total calories burned for the day.

4.3 Fitbit_Sleep

Field	Description
Id	User identifier. Foreign key to Fitbit_Users.Id.
SleepDay	Calendar date of the sleep record.
TotalSleepRecords	Count of separate sleep sessions logged on that date, e.g. a nap plus overnight sleep.
TotalMinutesAsleep	Total minutes spent asleep across all sessions that day.
TotalTimeInBed	Total minutes spent in bed across all sessions that day, including time awake.

4.4 Fitbit_Steps

Field	Description
Id	User identifier. Foreign key to Fitbit_Users.Id.
Steps Day	Calendar date of the step record.
Steps Hour	Hour of the day (0 to 23) the step count applies to.
No of Steps	Step count recorded in that hour. Absent for hours with no logged activity.

4.5 Fitbit_HR

Field	Description
Id	User identifier. Foreign key to Fitbit_Users.Id.
HR Date	Calendar date of the heart rate reading.
Hour	Hour of the day the reading was taken.
Minute	Minute within the hour the reading was taken.
HR	Heart rate reading, in beats per minute.